RUMMAN AHMED PRODHAN

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RESEARCH INTERESTS

Core Areas: Deep Learning, Biomedical Signal Processing (EEG/Audio), Federated Learning, Health Informatics.
Focus: Privacy-Preserving AI, Multimodal Fusion, Disentangled Representation Learning, XAI.

EDUCATION

BRAC University, Bangladesh *Status: Coursework Completed - Thesis Ongoing*
MSc. in Computer Science & Engineering CGPA: **3.95/4.00**
Advanced Coursework: Neural Networks & Fuzzy Systems, Distributed Computing, Deep Learning for Computer Vision, Symbolic Machine Learning, Pattern Recognition, Parallel Algorithms.
Awarded Merit Scholarship

University of Asia Pacific, Bangladesh *Apr 2018 - Jun 2022*
BSc. in Computer Science & Engineering
Cumulative CGPA: **3.79/4.00** *(Last 2 Years: **3.96**, Final Sem: **4.00**)*
Honors: Vice Chancellor's Honor List (Top 5% of Class), Dean's Honor List.

ACADEMIC & RESEARCH EXPERIENCE

Northern University Bangladesh *Sep 2022 - Present*
Senior Lecturer, CSE (Oct 2026 - Present) | *Lecturer, CSE* (Sep 2022 - Sep 2026)
Conducting BSc courses including Database Systems, AI, Software Development, Computer Architecture and Operating Systems.
Research Supervision: Currently supervising multiple undergraduate thesis groups focusing on machine learning and deep learning. Balanced full-time teaching with active research, resulting in multiple publications.

Neuro-Analytics Lab, NJIT, USA *Jul 2022 - June 2023*
Volunteer Assistant Researcher (Remote)
Contributed to EEG data analysis and emotion recognition research. Implemented feature extraction pipelines for neural activity data, resulting in an MDPI Sensors journal publication.

Bondstein Technologies Limited, Dhaka *Jan 2022 - Jun 2022*
Intern at Research & Development
Developed a real-time conveyor belt fault detection system for British American Tobacco (BAT) utilizing Raspberry Pi and Computer Vision techniques for industrial automation.

PUBLICATIONS & MANUSCRIPTS

[Conference] Abdullah Al Mobassir, Md. Rayhan Hasan, Mst. Afsana Sadia Borni, Md. Razu Ahmed, **Rumman Ahmed Prodhan**. *"Multimodal Deep Learning for Handwritten Prescription Recognition with Explainable AI." 2nd IEEE Quantum-inspired Patient Informatics (Q-PAIN)*, 2026. **(IEEE Xplore)**. doi:10.1109/QPAIN69676.2026.11546070

[Conference] Md Ohiduzzaman Ovi, Maliha Sanjana, Fahad, Mahjabin Runa, Zarin Tasnim Rothy, Tanmoy Sarkar Pias, A.M. Tayeful Islam, **Rumman Ahmed Prodhan**. *"Enhanced Pediatric Dental Segmentation Using a Custom SegUNet with VGG19 Backbone on Panoramic Radiographs"*. In *Proc. of 27th Int. Conf. on Computer and Information Technology (ICCIT)*, 2024. **(IEEE Xplore)**. doi:10.1109/ICCIT64611.2024.11021764

[Book Chapter] Rumman Ahmed Prodhan, Sumya Akter, Tanmoy Sarkar Pias, Md. Akhtaruzzaman Adnan. *"Optimal EEG Electrode Set for Emotion Recognition From Brain Signals: An Empirical Quest."* In *Human Activity and Behavior Analysis*, CRC Press, 2024. **(Taylor & Francis Group)**. doi:10.1201/9781003371540-5

[Journal] Sumya Akter[†], **Rumman Ahmed Prodhan**[†], Tanmoy Sarkar Pias, David Eisenberg, Jorge Fresneda Fernandez. *"M1M2: Deep Learning-Based Real-Time Emotion Recognition from Neural Activity."* Published in *Sensors*, Vol. 22, No. 21, 2022. **[Impact Factor: 3.9]**. doi:10.3390/s22218467 ([†]Equal Contribution).

[Conference] Rumman Ahmed Prodhan, Sumya Akter, Muhammad Bin Mujib, Md. Akhtaruzzaman Adnan, Tanmoy Sarkar Pias. *"Emotion Recognition from Brain Wave using Multitask Machine Learning Leveraging Residual Connections."* In *Machine Intelligence and Emerging Technologies (MIET)*, 2022. **(Springer)**. doi:10.1007/978-3-031-34622-4_10

[Conference] Sumya Akter, **Rumman Ahmed Prodhan**, Muhammad Bin Mujib, Md. Akhtaruzzaman Adnan, Tanmoy Sarkar Pias. *"Evaluating the Effectiveness of Classification Algorithms for EEG Sentiment Analysis."* In *Int. Conf. on Sentiment Analysis and Deep Learning (ICSADL)*, 2022. **(Springer)**. doi:10.1007/978-981-19-5443-6_17

MSC RESEARCH PROJECTS

Unsupervised Acoustic Disentanglement via Hybrid Beta-VAE
Addressed cross-cultural genre overlaps by fusing Convolutional Spectral Features (Audio) with Semantic Embeddings. Achieved **perfect separation (ARI=1.0)** on the HBLM-100 dataset. [Github]

Scalable Hybrid Parallelism for Federated Learning
Designed a high-performance architecture combining **MPI** (Inter-node) and **CUDA** (Intra-node) to solve the "Straggler Problem" in FL. Achieved a **9.26x speedup** over serial execution on the SipakMed dataset.

Federated Domain Generalization (FedDG) for Cancer Diagnosis
Implemented **Federated Batch Normalization (FedBN)** within the Flower framework to mitigate domain shifts between staining protocols. Improved cross-site robustness by 15% compared to standard FedAvg.

Zero-Shot Cross-Dataset EEG Emotion Recognition
Investigating domain generalization techniques (CORAL regularization, MixStyle) to transfer emotion recognition models between DEAP and DREAMER datasets without retraining.

LANGUAGE PROFICIENCY

IELTS Academic (Apr 2024): Overall Band: **8.0** (Listening: 8.5, Reading: 8.0, Writing: 7.0, Speaking: 7.5)

TECHNICAL SKILLS

Deep Learning: PyTorch, TensorFlow, Keras, Flower (FL)
Signal Processing: EEG Pipeline, FFT
Languages: Python, C++, SQL, MATLAB, Java
Parallel Computing: CUDA, MPI, OpenMP
Hardware: Raspberry Pi, Jetson Nano, Arduino
Web: Django, Bootstrap, HTML/CSS
Tools: Docker, Git, LaTeX, OpenCV, VOSviewer
Certification: MLOps Concepts (DataCamp)

REFERENCES

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